

WHITE BOX TESTING ASSINGMENT:

1.what are the advantages & disadvantages of white box testing?

Advantages of White box Testing:

White box testing provides deep insight into the internal structure of an application, improving code quality, reliability, and performance.

Thorough Code Coverage: Ensures detailed testing of internal logic, control structures, and execution paths to reduce untested or hidden code.

Code Optimization: Helps identify redundant, unused, or inefficient code to improve performance and maintainability.

Early Defect Detection: Enables early testing in development to quickly identify and fix defects before they become complex.

SDLC Easy Integration: Can be easily integrated into different phases of SDLC to support continuous testing and improvement.

Detection of Complex Defects: Helps uncover critical and hidden issues in code that may not be found through other testing methods.

The main disadvantages of white box testing include:

High Expertise Required:

Testers must possess a deep understanding of programming languages, internal architectures, and algorithms. It is generally unfeasible for non-technical or end-user testers to execute.

Time and Resource Intensive:

Evaluating every line of code, branch, and execution path takes substantial time, which significantly increases costs and delays for massive enterprise applications.

High Maintenance Overhead:

Every time the implementation details of the software change, the corresponding test cases often break and must be rewritten or updated, creating significant ongoing maintenance work.

Inability to Detect Missing Requirements:

White box testing evaluates the code that exists, but it cannot determine whether the software fulfills the original business requirements or if features are missing from the user's perspective.

Lack of User Perspective: Because developers and technical testers are intimately familiar with the internal logic, they may overlook usability issues, functional gaps, or user-facing bugs that would be immediately obvious to a standard end user.

2.what is statement coverage and path coverage?

Statement Coverage Testing:

Statement Coverage is a white-box testing technique used to verify that every executable statement in a program is executed at least once during testing. It helps ensure that the written code is properly tested and improves software reliability.

- Executes all program statements at least once
- Helps identify untested code
- Improves software quality and reliability

Path coverage: it is a white-box testing technique in software engineering that ensures every possible execution sequence (or path) through a piece of code is tested at least once, from start finish.

The Mechanic:

Path coverage is measured as the percentage of unique, independent paths executed by test cases relative to the total number of possible paths.